

Mathematical Foundations, Architecture Diagrams, and State Analysis of Modern LLM Training Methods — Part 3: Data Generation, Curation, Distillation, Compression, Agentic, Multimodal, Knowledge Injection, and Deployment

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Part I

Synthetic Data Generation

Self-Instruct

Mathematical Foundation

Self-Instruct bootstraps an instruction dataset from a small seed pool $\mathcal{S}$ by using the LLM itself as a generator. At each iteration, k seed instructions are sampled and the model generates a new instruction $\hat{x}$, its input, and output:

$$(\hat{x}, \hat{i}, \hat{y}) \sim P_{\theta}(\cdot \mid \text{few-shot}(\mathcal{S})) \quad (1)$$

Generated examples pass a quality filter (ROUGE similarity $< \delta$ to existing pool, length check):

$$\mathcal{S} \leftarrow \mathcal{S} \cup \{(\hat{x}, \hat{i}, \hat{y})\} \text{ iff } \max_{s \in \mathcal{S}} \text{ROUGE}(\hat{x}, s) < \delta \quad (2)$$

The growing pool is used to fine-tune θ , improving the quality of subsequent generations.

Architecture Flow Diagram

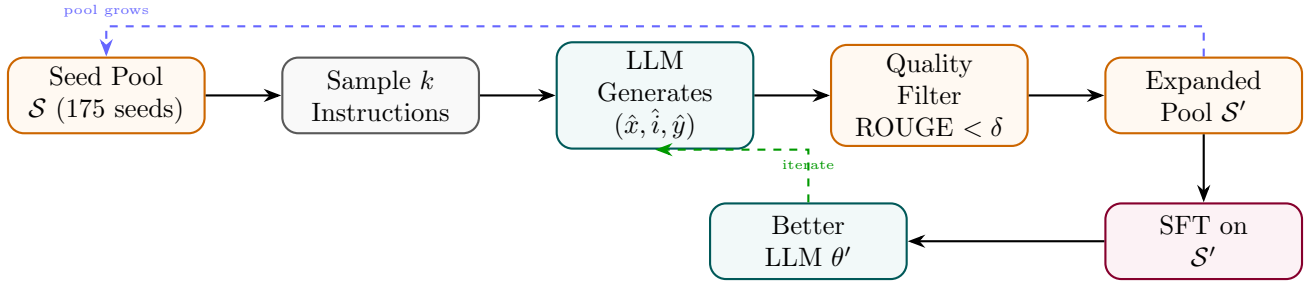


Figure 1: Self-Instruct: LLM generates new instructions from seed pool; quality filter expands the pool for SFT.

State Transition Table

Property	Before	After
Pool size	175 seed instructions	52,000+ generated
Annotation source	Human-written seeds	LLM-generated
Diversity filter	N/A	ROUGE deduplication
Training data cost	Low (seeds only)	Essentially free to scale
Instruction following	Weak	Strong

Table 1: State properties before and after Self-Instruct data generation.

Evol-Instruct

Mathematical Foundation

Evol-Instruct evolves existing instructions into harder variants via two mutation strategies. *In-depth* evolution adds constraints, increases steps, or requires deeper reasoning. *In-breadth* evolution creates new topic-adjacent instructions:

$$\hat{x} = M_{\text{depth}}(x) \text{ or } \hat{x} = M_{\text{breadth}}(x), \quad \hat{x} \sim P_{\text{teacher}}(\cdot \mid \text{evolve-prompt}, x) \quad (3)$$

Evolved instructions are validated (non-empty, no refusal, dissimilar from parent) and collected into $\mathcal{D}_{\text{evol}}$. Difficulty increases monotonically with evolution rounds E :

$$\mathbb{E}[\text{difficulty}(\hat{x}_E)] > \mathbb{E}[\text{difficulty}(\hat{x}_{E-1})] \quad (4)$$

Architecture Flow Diagram

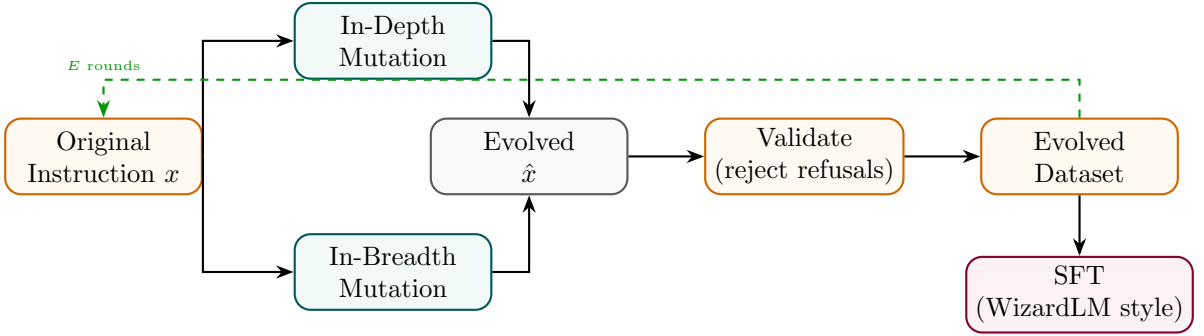


Figure 2: Evol-Instruct: in-depth and in-breadth mutations generate harder variants; validated pool used for SFT.

State Transition Table

Property	Before	After
Instruction difficulty	Low (seed)	Progressively higher (per round E)
Instruction diversity	Limited	Broad via breadth mutations
Data source	Human-written	LLM-evolved
Evolution rounds	0	E (typically 4–8)
Model performance	Baseline	WizardLM-style improvements

Table 2: State properties before and after Evol-Instruct.

Evol-Code

Mathematical Foundation

Evol-Code applies the same evolutionary principle to programming tasks, using code-specific mutation operators: adding input/output constraints, combining two functions, increasing time/space complexity, or switching programming language:

$$\hat{p} = M_{\text{code}}(p), \quad M \in \{\text{constrain, combine, optimize, translate}\} \quad (5)$$

Each evolved problem $\hat{p}$ is solved by a teacher model, and the (problem, solution) pair is validated by unit-test execution:

$$(\hat{p}, \hat{s}) \in \mathcal{D}_{\text{code}} \iff \text{unit_tests}(\hat{s}, \hat{p}) = \text{PASS} \quad (6)$$

Architecture Flow Diagram

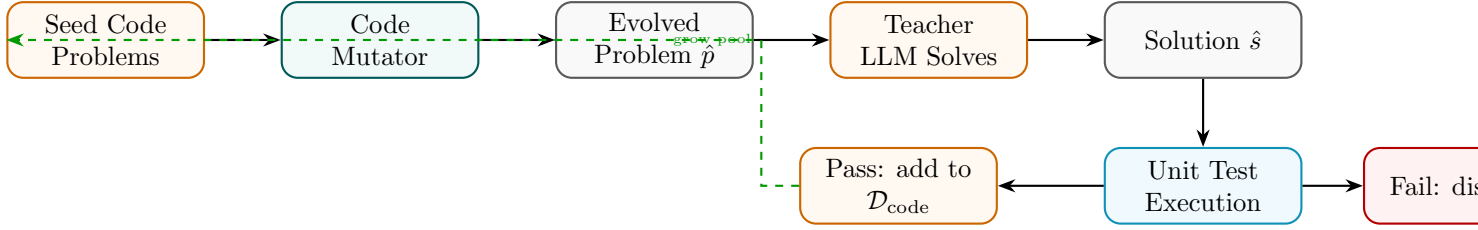


Figure 3: Evol-Code: code problems are mutated, solved by teacher, validated by unit tests before dataset inclusion.

State Transition Table

Property	Before	After
Problem complexity	Simple seed tasks	Multi-constraint, optimised
Validation method	Human review	Automated unit tests
Data quality gate	None	Unit-test pass rate
Programming languages	Fixed	Multi-language via translate mutation
Code reasoning ability	Baseline	Significantly improved

Table 3: State properties before and after Evol-Code data generation.

Synthetic Data Generation

Mathematical Foundation

A powerful teacher model P_T generates synthetic (input, output) pairs from topic/schema prompts, which are then used to train a smaller student P_S . The synthetic data distribution approximates the real task distribution P_{real} :

$$\mathcal{D}_{\text{synth}} = \{(x_i, y_i) : x_i \sim P_{\text{topic}}, y_i \sim P_T(\cdot | x_i)\} \quad (7)$$

Quality is improved by filtering on a scoring model or perplexity threshold τ_p :

$$(x, y) \in \mathcal{D}_{\text{final}} \iff \text{score}(x, y) \geq \tau_s \text{ and } \text{PPL}(y) \leq \tau_p \quad (8)$$

Architecture Flow Diagram

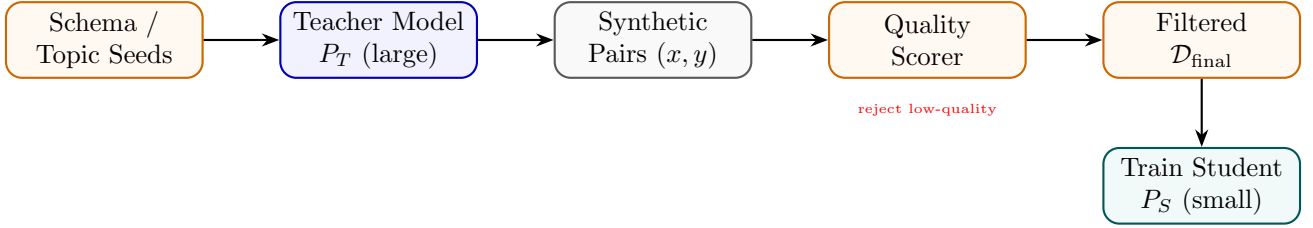


Figure 4: Synthetic Data Generation: teacher generates pairs from schema seeds; quality filter builds training set for student.

State Transition Table

Property	Before	After
Data source	Human annotation	LLM-generated at scale
Annotation cost	High	Very low (API cost)
Data volume	Thousands	Millions scalable
Quality control	Human review	Automated scoring
Domain coverage	Limited	Any topic via schema

Table 4: State properties before and after synthetic data generation.

Rejection Sampling

Mathematical Foundation

Rejection Sampling generates N candidate completions from the current policy and keeps only those that pass a quality threshold, building a filtered fine-tuning corpus:

$$\mathcal{D}_{\text{RS}} = \{(x, y^{(k)}) : y^{(k)} \sim \pi_{\theta}(x), r(x, y^{(k)}) \geq \tau, k = 1..N\} \quad (9)$$

Expected acceptance rate at threshold τ : $\alpha = P_{y \sim \pi_{\theta}}[r(x, y) \geq \tau]$. After SFT on $\mathcal{D}_{\text{RS}}$, the model’s probability mass shifts toward high-reward completions:

$$\mathbb{E}_{y \sim \pi_{\theta'}}[r(x, y)] > \mathbb{E}_{y \sim \pi_{\theta}}[r(x, y)] \quad (10)$$

Architecture Flow Diagram

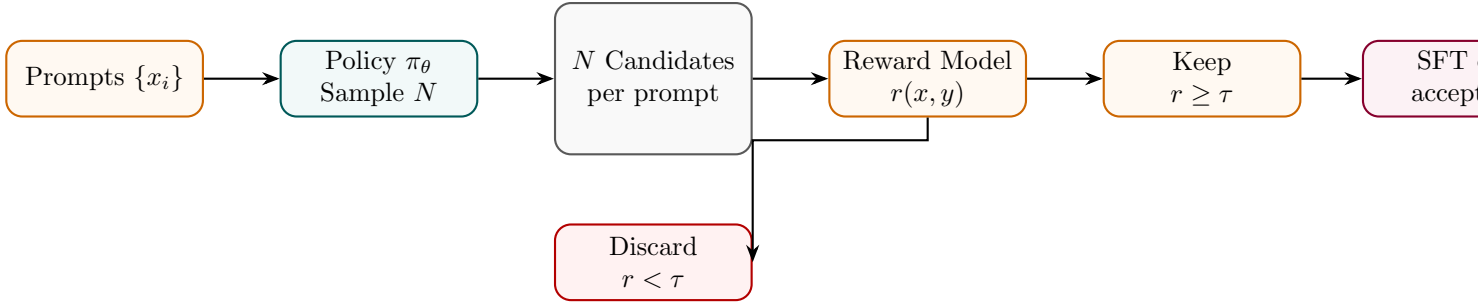


Figure 5: Rejection Sampling: N completions generated; only those above reward threshold kept for SFT.

State Transition Table

Property	Before	After
Training data quality	Mixed	High (above threshold τ)
Samples per prompt	1	N (with αN accepted)
Reward model	Optional	Required as filter
Iteration benefit	None	Compound with model updates
RL complexity	None	None (SFT only)

Table 5: State properties before and after Rejection Sampling.

Best-of-N Sampling

Mathematical Foundation

Best-of-N (BoN) sampling selects the highest-reward completion from N independent samples at inference time:

$$y^* = \arg \max_{k \in 1..N} r(x, y^{(k)}), \quad y^{(k)} \sim \pi_{\theta}(\cdot | x) \quad (11)$$

The expected reward improvement over single-sample baseline scales as:

$$\mathbb{E}[\max_k r(x, y^{(k)})] \approx \mu_r + \sigma_r \cdot \Phi^{-1}\left(\frac{N}{N+1}\right) \quad (12)$$

where μ_r, σ_r are the reward mean and std under π_{θ} , and Φ^{-1} is the inverse normal CDF. BoN can also be used to build SFT datasets by collecting (x, y^*) pairs.

Architecture Flow Diagram

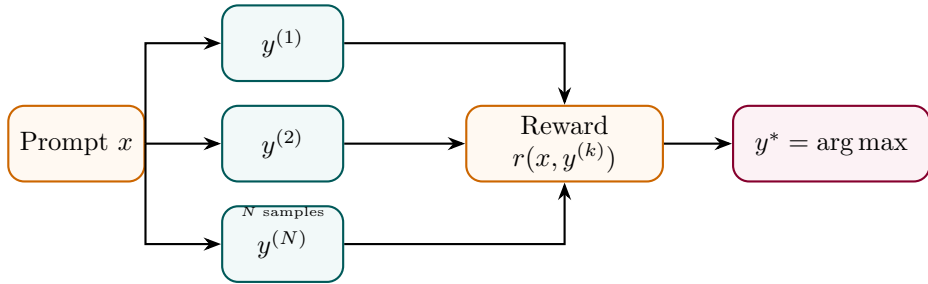


Figure 6: Best-of-N: N samples scored by reward model; highest reward selected as final output.

State Transition Table

Property	Before (single)	After (BoN)
Samples per query	1	N
Compute at inference	$1 \times$	$N \times$
Expected reward	μ_r	$\mu_r + O(\sigma_r \log N)$
Model weights	Unchanged	Unchanged
Best usage	Cheap inference	Quality-critical generation

Table 6: State properties comparing single-sample and Best-of-N sampling.

Part II

Data Curation

Data Filtering

Mathematical Foundation

Data filtering removes low-quality examples from a raw corpus using a set of quality signals $\{q_1, \dots, q_M\}$. A sample (x, y) is accepted iff it passes all signal thresholds:

$$(x, y) \in \mathcal{D}_{\text{clean}} \iff \bigwedge_{m=1}^M q_m(x, y) \geq \tau_m \tag{13}$$

Common signals include perplexity under a small LM (detects gibberish), toxic content classifiers, length ratios, and language identification. A learned quality classifier c_ϕ can consolidate multiple signals:

$$c_\phi(x, y) = \sigma(w^\top [q_1(x, y), \dots, q_M(x, y)]) \tag{14}$$

Architecture Flow Diagram

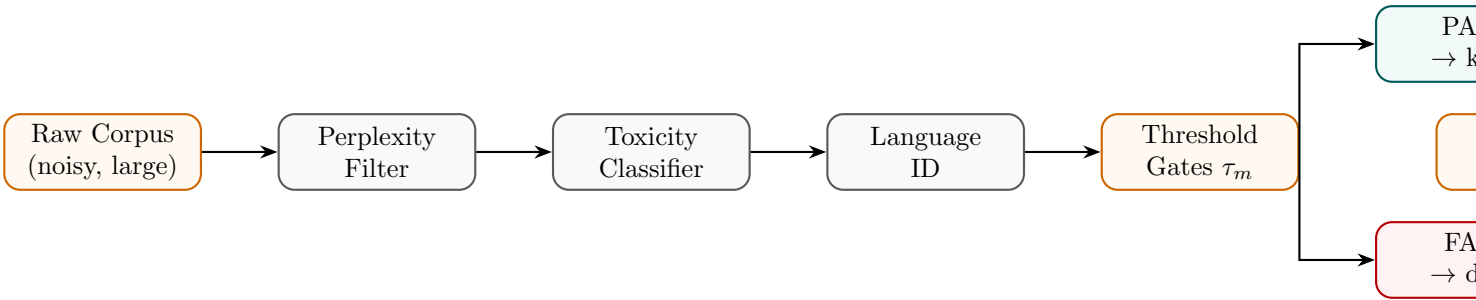


Figure 7: Data filtering pipeline: multiple quality signals applied in sequence; only passing samples retained.

State Transition Table

Property	Before	After
Corpus size	Very large (noisy)	Smaller (high quality)
Noise level	High	Low
Domain coverage	Broad (noisy)	Focused (clean)
Filter signals	None	M quality signals
Training efficiency	Low	Higher (less wasted compute)

Table 7: State properties before and after data filtering.

Deduplication

Mathematical Foundation

Deduplication removes near-duplicate documents to prevent memorisation and improve sample diversity. MinHash LSH estimates Jaccard similarity between documents:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \approx \frac{1}{K} \sum_{k=1}^K \mathbf{1}[h_k(A) = h_k(B)] \quad (15)$$

where $\{h_k\}$ are K independent hash functions and $h_k(A) = \min_{w \in \text{ngrams}(A)} h_k(w)$. Documents with $J(A, B) > \tau_J$ are considered duplicates and one is removed:

$$\mathcal{D}_{\text{dedup}} = \{d_i : \forall j < i, J(d_i, d_j) \leq \tau_J\} \quad (16)$$

Architecture Flow Diagram

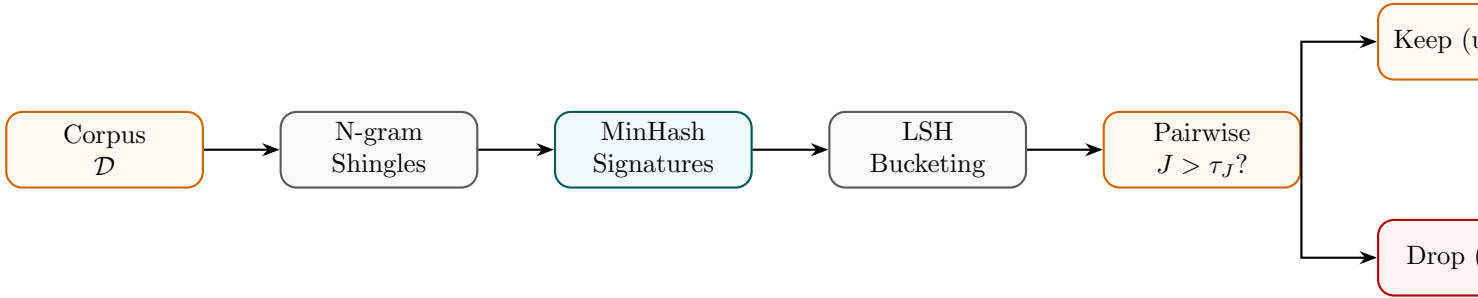


Figure 8: Deduplication: MinHash signatures group near-duplicates via LSH; high-similarity pairs discarded.

State Transition Table

Property	Before	After
Near-duplicate rate	15–30% (web data)	Near zero
Memorisation risk	High	Reduced
Effective token diversity	Low	High
Corpus size	$ \mathcal{D} $	$ \mathcal{D}_{\text{dedup}} < \mathcal{D} $
Compute wasted on duplicates	High	Eliminated

Table 8: State properties before and after deduplication.

Active Learning

Mathematical Foundation

Active Learning selects the most *informative* unlabelled examples for annotation, maximising model improvement per annotation dollar. Uncertainty sampling selects x^* with highest predictive entropy:

$$x^* = \arg \max_{x \in \mathcal{U}} \mathbb{H}[P_\theta(y | x)] = \arg \max_x - \sum_y P_\theta(y|x) \log P_\theta(y|x) \quad (17)$$

Query-by-committee uses disagreement between an ensemble $\{\theta_1, \dots, \theta_C\}$:

$$x^* = \arg \max_{x \in \mathcal{U}} \frac{1}{C} \sum_c \mathbb{KL}[P_{\theta_c}(\cdot|x) \| \bar{P}(\cdot|x)] \quad (18)$$

Architecture Flow Diagram

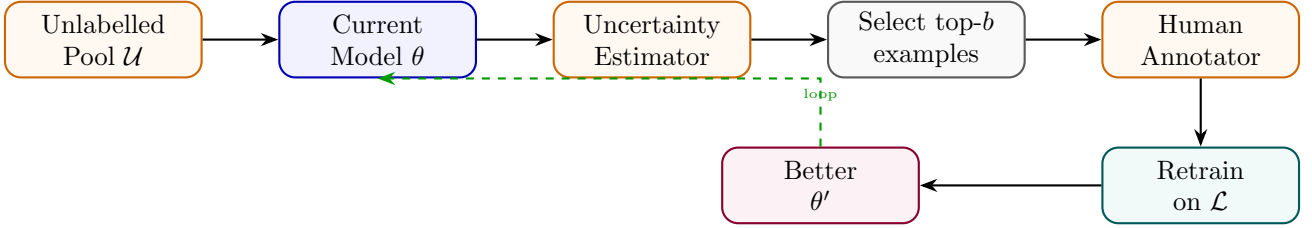


Figure 9: Active Learning: uncertainty estimator selects most informative examples; human annotates; model retrains.

State Transition Table

Property	Before	After
Annotation strategy	Random sampling	Uncertainty-driven
Labels per accuracy gain	Many	Few (targeted)
Unlabelled pool	Fully unused	Iteratively queried
Annotation budget	Fixed	Allocated by informativeness
Model improvement per label	Low	High

Table 9: State properties before and after Active Learning.

Curriculum Learning

Mathematical Foundation

Curriculum Learning trains on examples ordered by difficulty $d(x, y)$, starting with easy examples and progressively introducing harder ones. A competence function $c(t) \in [0, 1]$ controls what fraction of the difficulty range is accessible at step t :

$$c(t) = \min\left(1, c_0 + (1 - c_0) \frac{t}{T_{\text{ramp}}}\right) \quad (19)$$

The training distribution at step t only samples from examples with $d(x, y) \leq c(t) \cdot d_{\text{max}}$:

$$\mathcal{D}_t = \{(x, y) : d(x, y) \leq c(t) \cdot d_{\text{max}}\}, \quad (x, y) \sim \text{Uniform}(\mathcal{D}_t) \quad (20)$$

Architecture Flow Diagram

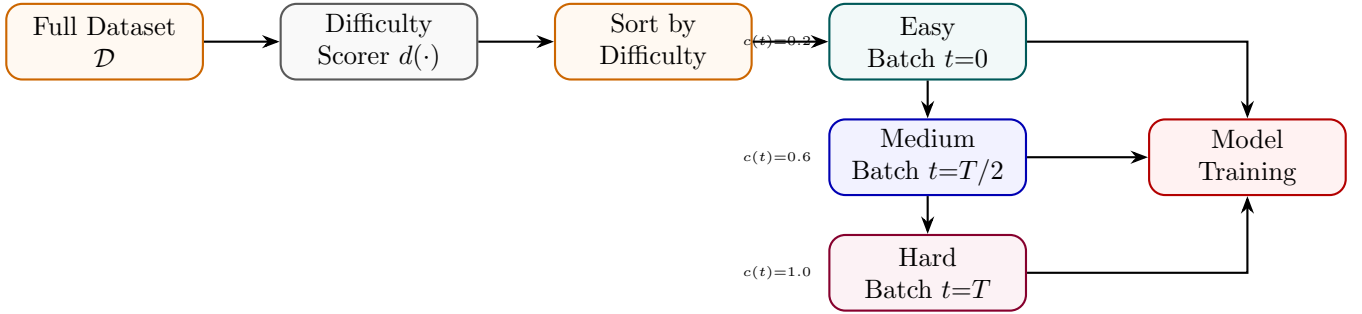


Figure 10: Curriculum Learning: examples sorted by difficulty; competence function progressively unlocks harder batches.

State Transition Table

Property	Before	After
Training order	Random	Difficulty-ordered
Early training batches	Mixed difficulty	Easy only
Convergence speed	Slower	Faster (easier start)
Final performance	Baseline	Improved
Difficulty metric	N/A	Perplexity, length, or task-specific

Table 10: State properties before and after Curriculum Learning.

Part III

Knowledge Distillation

Knowledge Distillation

Mathematical Foundation

Knowledge distillation transfers dark knowledge from a large teacher model f_{teacher} to a compact student model f_{student} . Let z_s and z_t be the logits of the student and teacher models respectively. The soft probability distribution at temperature T is computed as:

$$p_i(z, T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (21)$$

The total distillation loss is a weighted combination of hard target cross-entropy and soft target Kullback-Leibler (KL) divergence:

$$\mathcal{L}_{\text{KD}} = (1 - \alpha)\mathcal{L}_{\text{CE}}(y, p(z_s, 1)) + \alpha T^2 D_{\text{KL}}(p(z_t, T) \parallel p(z_s, T)) \quad (22)$$

where $\alpha \in [0, 1]$ balances the loss terms, and the temperature scaling factor T^2 scales the soft gradient updates to remain invariant of temperature changes:

$$\frac{\partial \mathcal{L}_{\text{KL}}}{\partial z_{s,i}} = \frac{1}{T} (p_i(z_s, T) - p_i(z_t, T)) \quad (23)$$

Architecture Flow Diagram

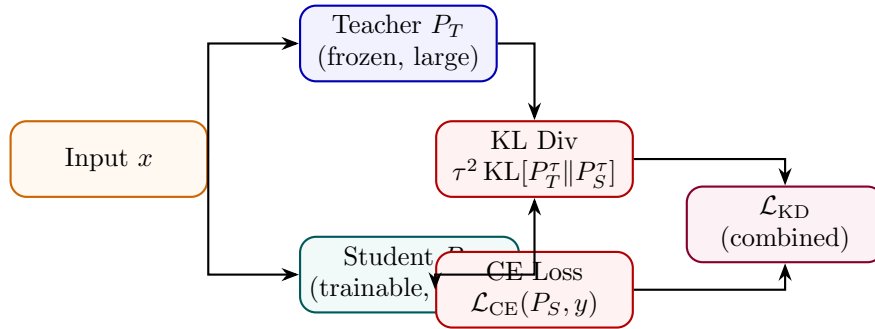


Figure 11: Knowledge Distillation: student learns from both hard labels (CE) and teacher soft logits (KL) simultaneously.

State Transition Table

Property	Before	After
Student accuracy	Low (random)	Near-teacher accuracy
Training signal	Hard labels only	Hard labels + soft teacher logits
Temperature τ	N/A	> 1 (typically 2–5)
Dark knowledge transfer	None	Via inter-class similarities
Student size	Large (teacher)	Small (~ 10 – $100\times$ fewer params)

Table 11: State properties before and after Knowledge Distillation.

Response Distillation

Mathematical Foundation

Response Distillation trains the student by directly imitating the teacher’s *full output sequence* rather than soft logits. The student minimises cross-entropy against teacher-generated responses:

$$\mathcal{L}_{\text{RD}} = - \sum_{(x, y_T) \in \mathcal{D}_T} \sum_t \log P_S(y_{T,t} \mid x, y_{T,<t}) \quad (24)$$

where $y_T \sim P_T(\cdot \mid x)$ are teacher completions. This is simpler than logit matching (no access to teacher logits required) and enables distillation from black-box APIs.

Architecture Flow Diagram

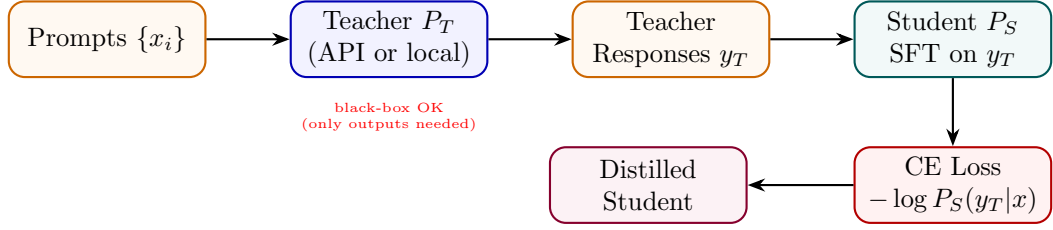


Figure 12: Response Distillation: teacher generates responses; student trained via SFT to imitate those outputs.

State Transition Table

Property	Before (KD)	After (Response KD)
Teacher access required	Logits (white-box)	Outputs only (black-box OK)
Loss type	KL on logits	CE on token sequence
Sequence-level knowledge	Partial	Full (complete generation)
Implementation complexity	High	Same as SFT
Applicable to proprietary APIs	No	Yes

Table 12: State properties comparing logit-level KD and Response Distillation.

Chain-of-Thought Distillation

Mathematical Foundation

CoT Distillation transfers the teacher’s *reasoning traces* to the student, so the student learns not just the answer but how to think step by step. The student is trained on (x, c_T, y_T) triplets where c_T is the teacher-generated chain:

$$\mathcal{L}_{\text{CoTD}} = - \sum_{(x, c_T, y_T)} \left[\sum_m \log P_S(c_{T,m} \mid x, c_{T,<m}) + \sum_t \log P_S(y_{T,t} \mid x, c_T, y_{T,<t}) \right] \quad (25)$$

This allows a small student to match the accuracy of a large teacher that reasons step by step.

Architecture Flow Diagram

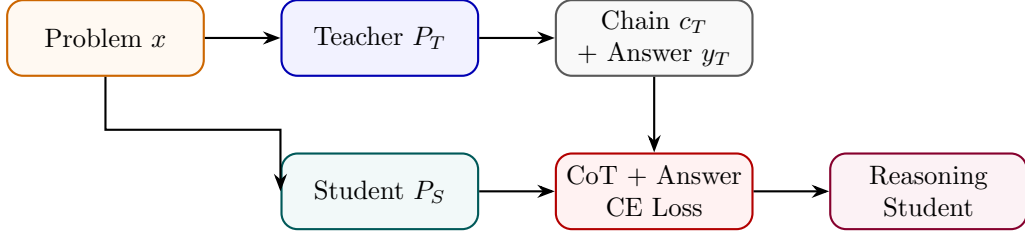


Figure 13: CoT Distillation: teacher’s reasoning chains and answers jointly train student to produce similar traces.

State Transition Table

Property	Before	After
Student reasoning	Direct answer	Step-by-step chain
Training target	Answer only	Chain + answer
Reasoning data source	None	Teacher-generated
Multi-step problem accuracy	Low	Significantly higher
Output verbosity	Terse	Explanatory

Table 13: State properties before and after CoT Distillation.

Policy Distillation

Mathematical Foundation

Policy Distillation minimises the KL divergence between a teacher policy π_T and student policy π_S over the full output distribution at every decoding step, not just the final token:

$$\mathcal{L}_{\text{PD}} = \mathbb{E}_{x \sim \mathcal{D}} \sum_t \text{KL}[\pi_T(\cdot | x, y_{<t}) || \pi_S(\cdot | x, y_{<t})] \quad (26)$$

Unlike response distillation (which uses sampled strings), policy distillation uses teacher token-level distributions and requires teacher logit access. This aligns the student’s generation *process* with the teacher’s, not just its outputs.

Architecture Flow Diagram

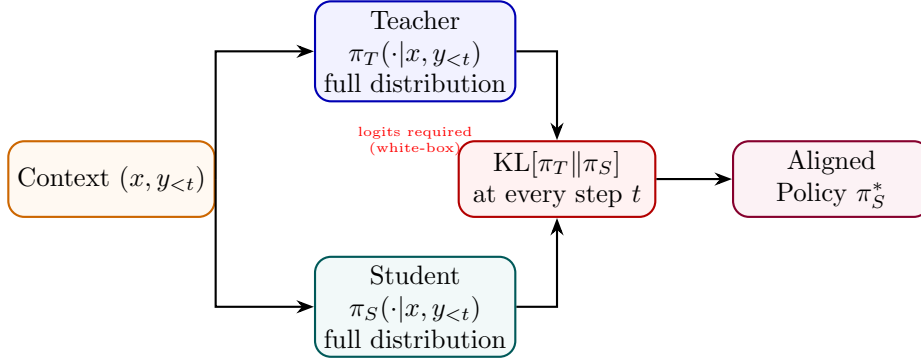


Figure 14: Policy Distillation: KL divergence between teacher and student distributions minimised at every token step.

State Transition Table

Property	Before (Response KD)	After (Policy KD)
Teacher signal	Sampled string	Full token distribution
KL divergence	N/A	Minimised per step
Teacher access	Black-box	White-box (logits)
Process alignment	Output only	Step-by-step
Overconfidence risk	Higher	Lower (distribution match)

Table 14: State properties comparing Response and Policy Distillation.

Reward Distillation

Mathematical Foundation

Reward Distillation compresses a large teacher reward model r_T into a smaller student reward model r_S by regression on teacher-generated scores:

$$\mathcal{L}_{\text{RewardD}} = \mathbb{E}_{(x,y)} [(r_T(x,y) - r_S(x,y))^2] \quad (27)$$

The student can also be trained with a ranking-consistency loss to preserve relative orderings even if absolute score values differ:

$$\mathcal{L}_{\text{rank}} = -\log \sigma(r_S(x, y_w) - r_S(x, y_l)) \text{ whenever } r_T(x, y_w) > r_T(x, y_l) \quad (28)$$

Architecture Flow Diagram

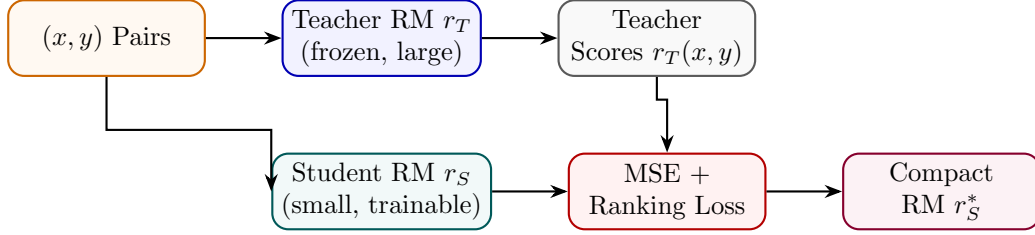


Figure 15: Reward Distillation: small student reward model trained via regression on large teacher reward scores.

State Transition Table

Property	Before	After
Reward model size	Large teacher	Small student
Inference latency	Slow	Fast
Score accuracy	High (teacher)	Near-teacher (distilled)
Training objective	Human preferences	Teacher score regression
Deployment cost	High	Low

Table 15: State properties before and after Reward Distillation.

Retrieval Distillation

Mathematical Foundation

Retrieval Distillation trains a student retriever q_S to match the relevance judgments of a larger teacher retriever q_T (or cross-encoder) without requiring access to teacher gradients. The student minimises the KL divergence between teacher and student retrieval distributions over a document set $\mathcal{C}$:

$$\mathcal{L}_{\text{RD}} = \mathbb{KL}[P_T(\cdot | q) \| P_S(\cdot | q)], \quad P(d | q) \propto \exp(q_\phi(q)^\top e(d)) \quad (29)$$

where q_ϕ is the query encoder and $e(d)$ is the document embedding.

Architecture Flow Diagram

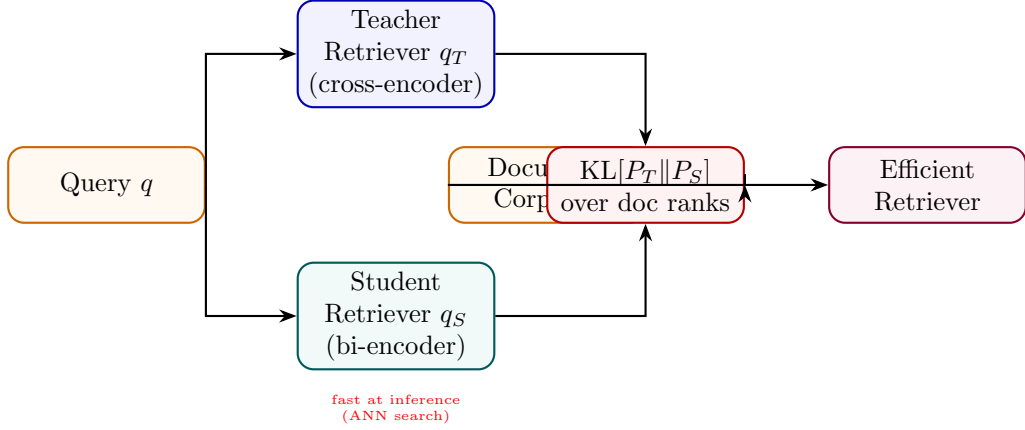


Figure 16: Retrieval Distillation: student bi-encoder trained to match teacher cross-encoder relevance distributions.

State Transition Table

Property	Before (Teacher)	After (Student)
Retriever type	Cross-encoder (slow)	Bi-encoder (fast ANN)
Query latency	$O(\mathcal{C})$	$O(\log \mathcal{C})$
Ranking quality	Very high	Near-teacher
Index requirement	None	Pre-computed embeddings
Training signal	Human judgments	Teacher soft scores

Table 16: State properties comparing teacher cross-encoder and student bi-encoder after retrieval distillation.

Part IV

Model Compression

Quantization-Aware Training (QAT)

Mathematical Foundation

QAT simulates low-bit precision during model training, allowing the optimizer to adjust parameters to compensate for quantization errors. Let w be the continuous weights. The quantized representation $\bar{w}$ is calculated as:

$$\bar{w} = S \cdot \text{clip} \left(\text{round} \left(\frac{w}{S} \right) - Z, q_{\min}, q_{\max} \right) \quad (30)$$

where S is the quantization scale factor, Z is the zero-point, and $[q_{\min}, q_{\max}]$ defines the integer boundary (e.g. $[-128, 127]$ for 8-bit quantization). Because the rounding function has zero gradients almost everywhere ($\frac{\partial \text{round}(x)}{\partial x} = 0$), standard backpropagation fails. QAT resolves this by applying the Straight-Through Estimator (STE) during the backward pass, approximating the gradient of the rounding operator as identity:

$$\frac{\partial \bar{w}}{\partial w} \approx \begin{cases} 1 & \text{if } w \in [w_{\min}, w_{\max}] \\ 0 & \text{otherwise} \end{cases} \quad (31)$$

Architecture Flow Diagram

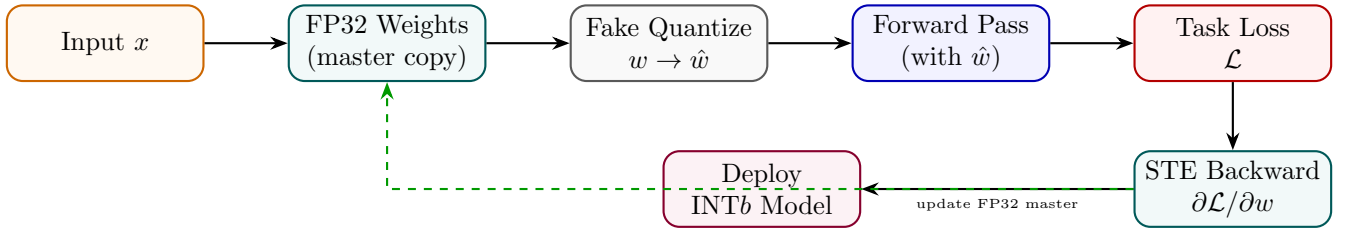


Figure 17: QAT: fake quantization in forward pass; STE allows gradients to flow to FP32 master weights.

State Transition Table

Property	Before (PTQ)	After (QAT)
Quantization awareness	None during training	Simulated in forward pass
Accuracy loss	1–5% degradation	<1% typical
Calibration data	Small set required	Full training data
Weight format	FP32	FP32 master, INTb deploy
Gradient flow	Normal	Via STE

Table 17: State properties comparing Post-Training Quantization and QAT.

Pruning

Mathematical Foundation

Pruning removes weights whose contribution to the model output is below a threshold, producing a sparse network. Magnitude-based unstructured pruning removes the fraction p of weights with the smallest absolute values:

$$m_i = \mathbf{1}[|w_i| \geq \tau_p], \quad W_{\text{sparse}} = W \odot m \quad (32)$$

where τ_p is chosen to achieve sparsity ratio $s = 1 - \|m\|_1/\|m\|_0$. Structured pruning removes entire heads or neurons (rows/columns) and is hardware-friendly:

$$\text{importance}(h) = \sum_{i \in h} |w_i| \cdot |\nabla_{w_i} \mathcal{L}| \quad (33)$$

Architecture Flow Diagram

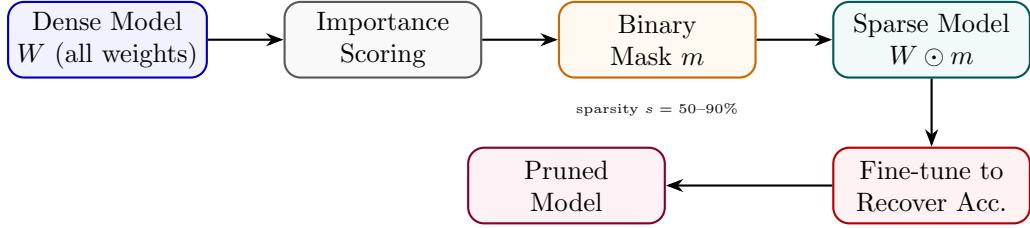


Figure 18: Pruning: importance scores determine binary mask; sparse model fine-tuned to recover accuracy.

State Transition Table

Property	Before	After
Model sparsity	0%	50–90%
Parameter count	$ \theta $	$(1 - s) \theta $
Inference FLOPs	Baseline	Reduced (structured)
Accuracy	Baseline	Slightly reduced
Memory footprint	Full	Reduced by s

Table 18: State properties before and after pruning.

Post-Training Quantization (PTQ)

Mathematical Foundation

PTQ converts a trained FP32 model to reduced precision (INT8, INT4) without additional training. Per-channel scale factors s_c are calibrated on a small unlabelled dataset $\mathcal{D}_{\text{cal}}$:

$$s_c = \frac{\max_{x \in \mathcal{D}_{\text{cal}}} |a_c(x)|}{2^{b-1} - 1}, \quad \hat{a}_c = \text{round}(a_c/s_c) \cdot s_c \quad (34)$$

GPTQ computes optimal quantization for each weight row by minimising the second-order reconstruction error using the Hessian:

$$\min_{\hat{W}} \|\hat{W}X - WX\|_F^2 \approx \sum_i (w_i - \hat{w}_i)^2 [H^{-1}]_{ii} \quad (35)$$

Architecture Flow Diagram

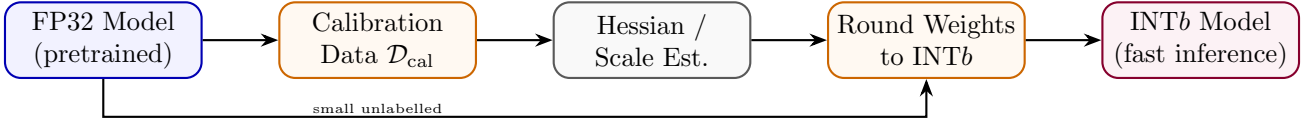


Figure 19: PTQ: calibration statistics guide per-channel scales; weights rounded to INT b without retraining.

State Transition Table

Property	Before	After
Weight precision	FP32 / BF16	INT8 / INT4
Memory usage	$4 \times n$ bytes	$1\text{--}0.5 \times n$ bytes
Retraining required	N/A	None (calibration only)
Accuracy loss	0%	0.5–3% (INT8), higher INT4
Calibration data	Not needed	Small sample (128–512 examples)

Table 19: State properties before and after Post-Training Quantization.

Part V

Agentic Training

Tool-Use Training

Mathematical Foundation

Tool-Use Training teaches the model to generate structured tool invocations interleaved with reasoning (ReAct-style). The training data consists of trajectories $\tau = (t_1, a_1, o_1, \dots, t_N, a_N, o_N, y)$ where t_i is a thought, a_i is a tool call, and o_i is the observation. The model is trained to predict thought and action tokens conditioned on prior context and observations:

$$\mathcal{L}_{\text{tool}} = - \sum_{i=1}^N \sum_j \log P_{\theta}(t_{i,j} \mid \text{context}_{<i}) + \log P_{\theta}(a_{i,j} \mid \text{context}_{<i}, t_i) \quad (36)$$

Architecture Flow Diagram

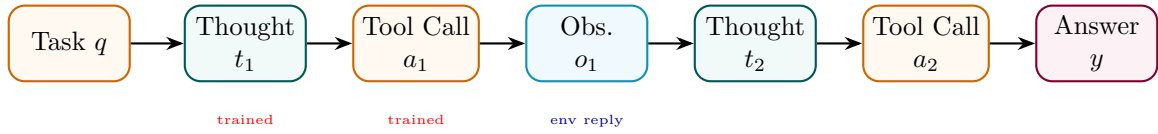


Figure 20: Tool-Use Training: thought-action-observation trajectory; model trained on thought and action tokens.

State Transition Table

Property	Before	After
Output format	Free text	Thought + tool call interleaved
Tool invocation	None	Structured API calls
Observation integration	N/A	Conditioned on tool results
Multi-step planning	Weak	Strong
Training data type	Static QA	Trajectory (thought, action, obs)

Table 20: State properties before and after Tool-Use Training.

Function Calling Training

Mathematical Foundation

Function Calling Training specifically trains the model to produce well-formed JSON function-call objects given a natural-language request and a tool schema. The model maps input x and schema $S = \{f_1, \dots, f_K\}$ to a structured call $c = \{\text{name: } f_k, \text{args: } \{a_1:v_1, \dots\}\}$:

$$\mathcal{L}_{\text{FC}} = -\log P_{\theta}(c \mid x, S) \tag{37}$$

where c is the ground-truth function call. Schema-aware token masking ensures the model only generates valid JSON that matches the function signature, reducing hallucination of non-existent arguments.

Architecture Flow Diagram

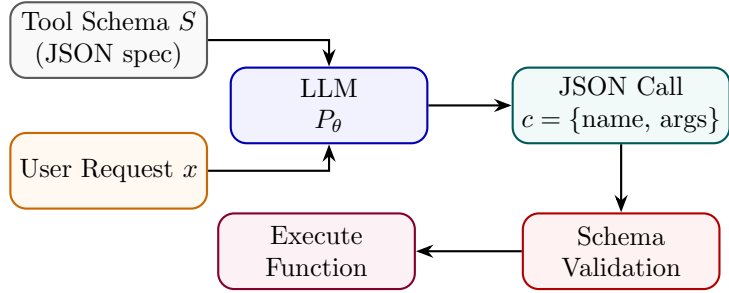


Figure 21: Function Calling Training: model receives request + schema, outputs structured JSON call validated against spec.

State Transition Table

Property	Before	After
Output format	Natural language	Structured JSON
Schema adherence	None	Enforced
Argument hallucination	High	Low
API integration	Manual parsing	Direct execution
Training data	Text completions	(request, schema, call) triplets

Table 21: State properties before and after Function Calling Training.

Multi-Agent Training

Mathematical Foundation

Multi-Agent Training trains a collection of specialised agents $\{\pi_1, \dots, \pi_K\}$ and an orchestrator π_0 to collaboratively solve complex tasks. The orchestrator decomposes a task T into sub-tasks $\{t_1, \dots, t_K\}$ and assigns them to specialists:

$$\pi_0 : T \rightarrow \{(t_k, k)\}_{k=1}^K, \quad \pi_k : t_k \rightarrow y_k, \quad y = \pi_0(\{y_k\}) \quad (38)$$

Training uses task-completion reward at the full-task level, distributed to agents via credit assignment:

$$r_k = r_{\text{task}} \cdot \mathbf{1}[\pi_k \text{ contributed to solution}] \quad (39)$$

Architecture Flow Diagram

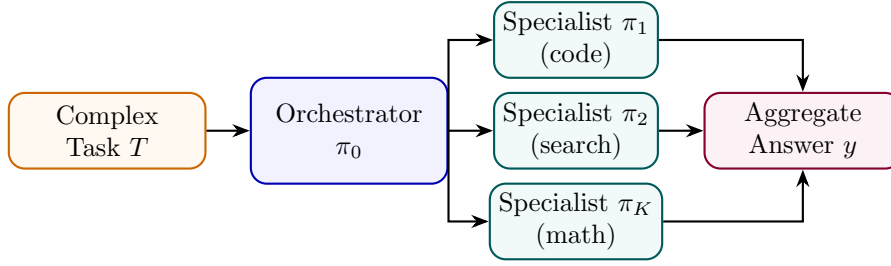


Figure 22: Multi-Agent Training: orchestrator decomposes task and assigns sub-tasks to specialised agents.

State Transition Table

Property	Before	After
Task handling	Single monolithic model	Orchestrator + specialists
Specialisation	None	Per-domain agents
Credit assignment	N/A	Per-agent contribution
Task complexity ceiling	Single-context limit	Decomposable to multiple contexts
Coordination	N/A	Learned delegation protocol

Table 22: State properties before and after Multi-Agent Training.

Part VI

Multimodal Training

Vision-Language Training

Mathematical Foundation

Vision-Language Training jointly trains a visual encoder f_v and language model f_l to process image-text pairs. A projection layer P aligns visual features to the LLM’s token embedding space:

$$v_{\text{tokens}} = P(f_v(I)), \quad \tilde{x} = [v_{\text{tokens}}; E(x_{\text{text}})] \tag{40}$$

Training uses a masked language modelling or auto-regressive loss on the text tokens conditioned on visual tokens:

$$\mathcal{L}_{\text{VL}} = - \sum_t \log P_{f_l}(x_t \mid v_{\text{tokens}}, x_{<t}) \tag{41}$$

Architecture Flow Diagram

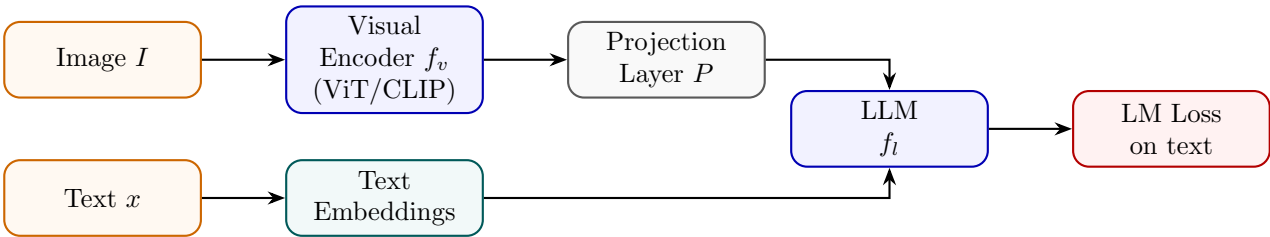


Figure 23: Vision-Language Training: visual encoder outputs projected to LLM token space; joint auto-regressive loss.

State Transition Table

Property	Before	After
Input modality	Text only	Text + images
Visual understanding	None	Strong (via encoder)
Projection layer	Not present	Trained to align modalities
Alignment quality	N/A	Measured by VQA, captioning
Training data	Text corpora	Image-caption / VQA pairs

Table 23: State properties before and after Vision-Language Training.

Grounding Training

Mathematical Foundation

Grounding Training teaches the model to link text spans to image regions (bounding boxes). Given a referring expression e and image I , the model predicts box coordinates (x_1, y_1, x_2, y_2) normalised to $[0, 1]^4$:

$$\hat{b} = f_{\theta}(I, e), \quad \mathcal{L}_{\text{GND}} = \text{L1}(\hat{b}, b^*) + \lambda(1 - \text{IoU}(\hat{b}, b^*)) \quad (42)$$

where b^* is the ground-truth box and IoU is intersection-over-union. The model generates boxes as text tokens (serialised coordinates), enabling unified seq2seq training.

Architecture Flow Diagram

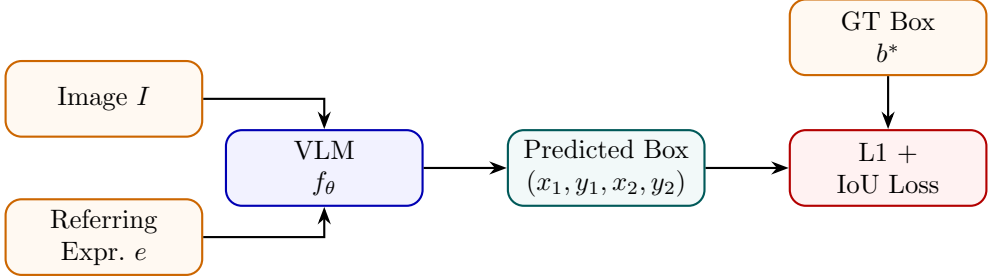


Figure 24: Grounding Training: VLM predicts bounding box coordinates from image and referring expression.

State Transition Table

Property	Before	After
Output type	Text description	Bounding box coordinates
Spatial understanding	Implicit	Explicit (pixel-level)
Loss type	Cross-entropy	L1 + IoU
Grounding ability	None	Text-to-region linking
Training data	Captions	(image, expression, bbox) triplets

Table 24: State properties before and after Grounding Training.

Part VII

Knowledge Injection

Retrieval-Augmented Training

Mathematical Foundation

Retrieval-Augmented Training (RAT) trains the model to use retrieved documents $\mathcal{R}(x) = \{d_1, \dots, d_K\}$ as context. The model generates y conditioned on the query and retrieved set:

$$P_\theta(y \mid x, \mathcal{R}(x)) = \prod_t P_\theta(y_t \mid x, d_1, \dots, d_K, y_{<t}) \quad (43)$$

Training includes both retrieval-present and retrieval-absent examples to prevent over-reliance on context. A closed-book loss component ensures the model retains parametric knowledge:

$$\mathcal{L}_{\text{RAT}} = \lambda \mathcal{L}_{\text{open-book}} + (1 - \lambda) \mathcal{L}_{\text{closed-book}} \quad (44)$$

Architecture Flow Diagram

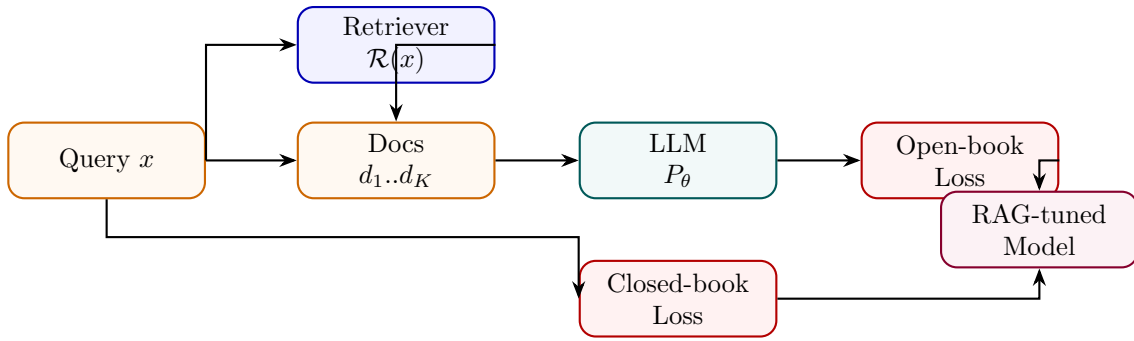


Figure 25: RAT: model trained on both retrieval-augmented and closed-book losses to balance open/parametric knowledge.

State Transition Table

Property	Before	After
Context awareness	Parametric only	Retrieved + parametric
Knowledge staleness	Fixed (training cutoff)	Dynamic (fresh retrieval)
Hallucination rate	Higher	Lower (grounded in docs)
Retriever dependency	None	Required at inference
Parametric retention	Full	Preserved (closed-book loss)

Table 25: State properties before and after Retrieval-Augmented Training.

Memory Tuning

Mathematical Foundation

Memory Tuning trains the model to store and retrieve episodic memories $\mathcal{M} = \{(k_i, v_i)\}$ in an external key-value store. The model learns to write memories via a write head w_θ and retrieve via dot-product attention:

$$\text{retrieve}(q) = \sum_i \text{softmax}(q^\top k_i / \sqrt{d}) \cdot v_i, \quad q = f_\theta^{\text{query}}(x) \quad (45)$$

Training uses a distractor task: memory-relevant queries should prefer correct memory entries, trained with a contrastive retrieval loss:

$$\mathcal{L}_{\text{mem}} = -\log \frac{\exp(q^\top k^+ / \tau)}{\sum_j \exp(q^\top k_j / \tau)} \quad (46)$$

Architecture Flow Diagram

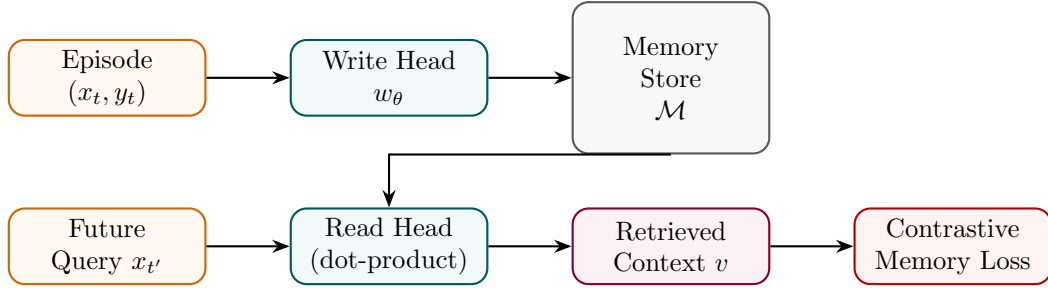


Figure 26: Memory Tuning: write head stores episodes; read head retrieves via attention; contrastive loss trains both.

State Transition Table

Property	Before	After
Memory type	Parametric only	Parametric + episodic KV store
Context window	Fixed T tokens	Unbounded (external memory)
Personalisation	None	Per-user episodic recall
Read/write mechanism	None	Learned attention-based
Long-term recall	Poor	Strong

Table 26: State properties before and after Memory Tuning.

Domain Knowledge Injection

Mathematical Foundation

Domain Knowledge Injection embeds structured domain knowledge (ontologies, knowledge graphs, fact databases) into the model. Entities e and relations r from a KG are embedded alongside token embeddings and injected via cross-attention:

$$h'_t = h_t + \text{CrossAttn}(h_t, \{e_i, r_{ij}\}_{(i,j) \in \mathcal{G}}) \quad (47)$$

Training uses a knowledge grounding loss that verifies generated text against retrieved facts:

$$\mathcal{L}_{\text{KI}} = \mathcal{L}_{\text{LM}} + \mu \mathcal{L}_{\text{fact-check}} \quad (48)$$

Architecture Flow Diagram

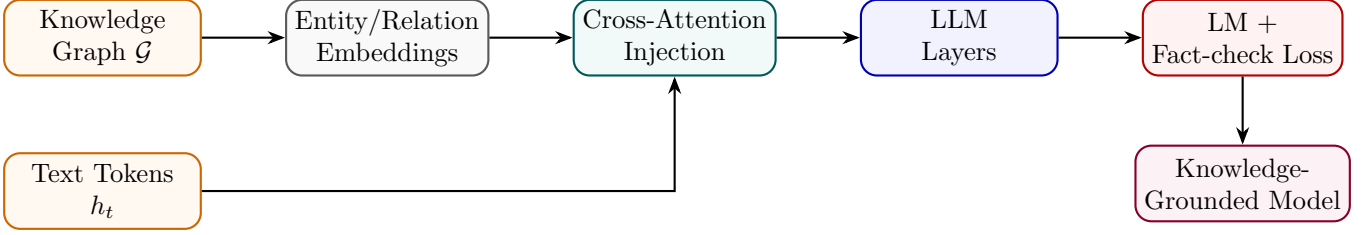


Figure 27: Domain Knowledge Injection: KG entity/relation embeddings injected into LLM via cross-attention.

State Transition Table

Property	Before	After
Knowledge source	Implicit (pretraining)	Explicit KG entities
Fact accuracy	Moderate	Significantly higher
Knowledge update	Retrain required	KG update sufficient
Cross-attention overhead	None	One extra module per layer
Domain coverage	General	Domain-specific + general

Table 27: State properties before and after Domain Knowledge Injection.

Part VIII

Deployment and Training Efficiency

Mixed Precision Training

Mathematical Foundation

Mixed Precision Training maintains FP32 master weights but computes forward and backward passes in FP16/BF16. A loss scaling factor S prevents underflow of small gradients:

$$\tilde{\mathcal{L}} = S \cdot \mathcal{L}, \quad g_{\text{FP32}} = \frac{1}{S} g_{\text{FP16}} \quad (49)$$

BF16 has the same exponent range as FP32 (8 bits) but fewer mantissa bits (7 vs 23), making it more numerically stable without loss scaling:

$$\text{memory}(\text{FP16}) = 2N \text{ bytes vs } \text{memory}(\text{FP32}) = 4N \text{ bytes} \quad (50)$$

Architecture Flow Diagram

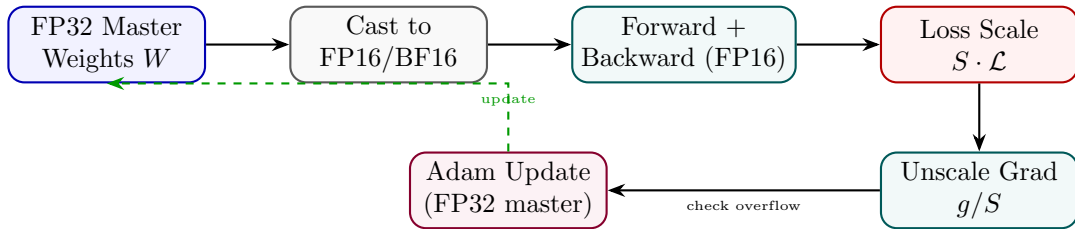


Figure 28: Mixed Precision Training: FP16 compute with FP32 master weights; loss scaling prevents gradient underflow.

State Transition Table

Property	Before (FP32)	After (Mixed FP16)
Compute precision	FP32	FP16/BF16
Memory footprint	4N bytes	2N + 4N (half + master)
Training speed	Baseline	2–3× faster
Numerical stability	High	Managed via loss scaling
GPU tensor cores	Unused	Utilised (FP16/BF16 native)

Table 28: State properties before and after Mixed Precision Training.

Gradient Checkpointing

Mathematical Foundation

Gradient Checkpointing trades compute for memory by not storing all intermediate activations during the forward pass. Instead, activations at checkpoint nodes $\mathcal{C} \subset \{1, \dots, L\}$ are stored; all others are recomputed during the backward pass:

$$\text{Memory}_{\text{GC}} = O(\sqrt{L} \cdot d), \quad \text{vs } O(L \cdot d) \text{ without GC} \quad (51)$$

For L layers, optimal checkpointing every $\sqrt{L}$ layers minimises memory while adding one extra forward pass overhead:

$$\text{Compute}_{\text{GC}} \approx 2 \cdot \text{Compute}_{\text{standard}} \quad (52)$$

Architecture Flow Diagram

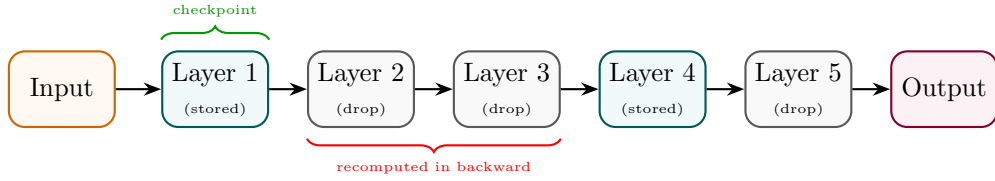


Figure 29: Gradient Checkpointing: only checkpoint activations stored; intermediate activations recomputed in backward.

State Transition Table

Property	Before	After
Activation memory	$O(L \cdot d)$	$O(\sqrt{L} \cdot d)$
Recomputation	None	Extra forward for non-checkpoints
Max batch size	Limited by activations	Larger (mem freed)
Compute overhead	Baseline	$\sim 30\%$ more FLOPs
Large model feasibility	Memory OOM	Enabled

Table 29: State properties before and after Gradient Checkpointing.

Flash Attention

Mathematical Foundation

FlashAttention accelerates attention computation by tiling input blocks in SRAM and utilizing online softmax reduction to avoid materializing the massive $N \times N$ attention matrix in HBM. Let Q, K, V be split into blocks Q_i, K_j, V_j of sizes B_c, B_r . The online softmax computes running scaling statistics. For a query block Q_i : At step j (processing key block K_j), we compute raw attention scores:

$$S^{(j)} = \frac{Q_i K_j^\top}{\sqrt{d}} \quad (53)$$

We update running row-wise maximums $m^{(j)}$ and denominators $d^{(j)}$:

$$\tilde{m}^{(j)} = \max(S^{(j)}) \quad (54)$$

$$m^{(j)} = \max(m^{(j-1)}, \tilde{m}^{(j)}) \quad (55)$$

$$\tilde{d}^{(j)} = \sum_k \exp(S_k^{(j)} - m^{(j)}) \quad (56)$$

$$d^{(j)} = d^{(j-1)} \exp(m^{(j-1)} - m^{(j)}) + \tilde{d}^{(j)} \quad (57)$$

The running attention output block O_i is updated dynamically using:

$$O_i^{(j)} = \frac{d^{(j-1)} \exp(m^{(j-1)} - m^{(j)}) \cdot O_i^{(j-1)} + \exp(\tilde{m}^{(j)} - m^{(j)}) \cdot \tilde{P}^{(j)} V_j}{d^{(j)}} \quad (58)$$

where $\tilde{P}^{(j)} = \exp(S^{(j)} - \tilde{m}^{(j)})$. This achieves exact attention outputs with $O(N)$ memory access instead of $O(N^2)$.

Architecture Flow Diagram

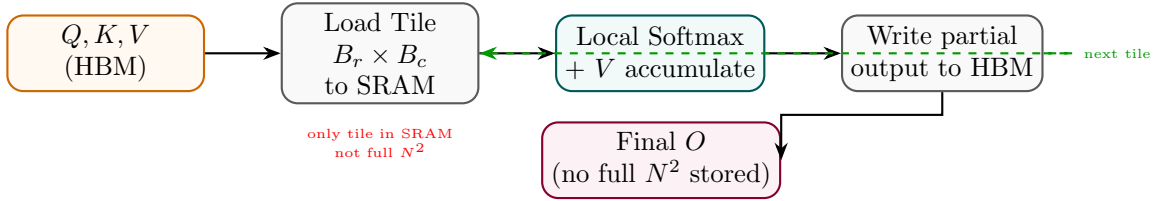


Figure 30: Flash Attention: tiled block-wise computation avoids materialising the full attention matrix in HBM.

State Transition Table

Property	Before (Standard)	After (Flash Attention)
Attention matrix memory	$O(N^2)$ HBM	$O(N)$ (tiles in SRAM)
HBM read/write passes	Multiple	Minimal (one pass)
Compute FLOPs	$O(N^2 d)$	$O(N^2 d)$ (same)
Speed (wall-clock)	Baseline	2–4× faster
Long-sequence feasibility	OOM at $N > 4096$	$N > 100,000$ feasible

Table 30: State properties comparing Standard and Flash Attention.

Fully Sharded Data Parallel (FSDP)

Mathematical Foundation

FSDP shards all three components of the model state (Parameters Ψ , Gradients G , and Optimizer States O) across all N available GPUs. Let the model parameters occupy $M_{\text{param}} = 2\Psi$ bytes in 16-bit precision, gradients occupy $M_{\text{grad}} = 2\Psi$ bytes, and Adam optimizer states occupy $M_{\text{opt}} = 12\Psi$ bytes (storing FP32 copies of weights, first moments, and second moments). Under standard Distributed Data Parallel (DDP), the parameters, gradients, and optimizer states are fully duplicated on each GPU:

$$M_{\text{baseline}} = 2\Psi + 2\Psi + 12\Psi = 16\Psi \text{ bytes} \quad (59)$$

FSDP shards these states across N GPUs. The peak memory occupancy per GPU is:

$$M_{\text{FSDP}} = \underbrace{\frac{2\Psi}{N}}_{\text{sharded params}} + \underbrace{\frac{2\Psi}{N}}_{\text{sharded grads}} + \underbrace{\frac{12\Psi}{N}}_{\text{sharded optimizer}} + \underbrace{2\Psi_{\text{block}}}_{\text{reconstructed block}} \quad (60)$$

During the forward pass, an ‘All-Gather’ operation reconstructs parameters block-by-block on the fly, which are discarded immediately after computing the block activation. During the backward pass, parameters are gathered again, gradients are computed, and a ‘Reduce-Scatter’ operation synchronizes and shards the gradients across GPUs.

Architecture Flow Diagram

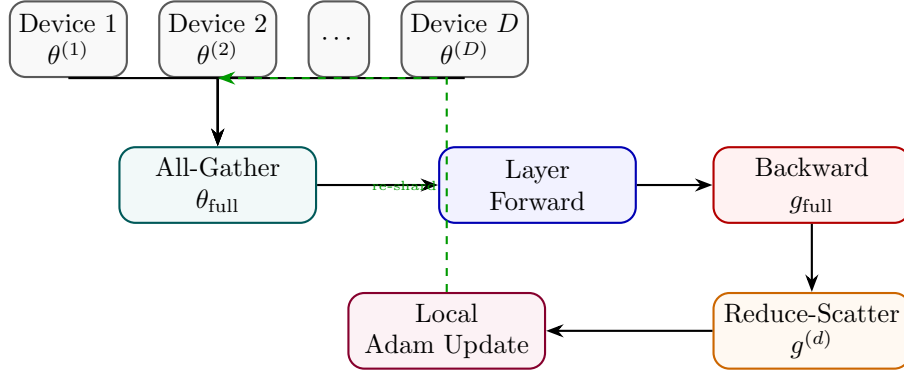


Figure 31: FSDP: parameters sharded across devices; All-Gather for compute, Reduce-Scatter for gradient aggregation.

State Transition Table

Property	Before (DDP)	After (FSDP)
Per-device param memory	$O(\theta)$	$O(\theta /D)$
Optimizer state memory	$O(\theta)$ per device	$O(\theta /D)$ per device
Communication pattern	AllReduce (gradients)	All-Gather + Reduce-Scatter
Max model size	GPU VRAM limited	$D\times$ larger
Training throughput	Baseline	Near-linear scaling

Table 31: State properties comparing DDP and FSDP.

DeepSpeed ZeRO

Mathematical Foundation

ZeRO (Zero Redundancy Optimizer) partitions the memory footprint of the model across data-parallel processes. It consists of three successive stages:

1. **ZeRO-1 (Optimizer State Partitioning)**: The optimizer states (e.g. Adam states) are sharded across N_{dp} data-parallel ranks. Parameters and gradients remain fully replicated. Peak memory per GPU is:

$$M_{\text{ZeRO-1}} = 2\Psi + 2\Psi + \frac{12\Psi}{N_{dp}} \quad (61)$$

2. **ZeRO-2 (Gradient Partitioning)**: Gradients are sharded immediately after their calculation in the backward pass. Peak memory per GPU is:

$$M_{\text{ZeRO-2}} = 2\Psi + \frac{2\Psi}{N_{dp}} + \frac{12\Psi}{N_{dp}} \quad (62)$$

3. **ZeRO-3 (Parameter Partitioning)**: Parameters are sharded across the ranks. The forward and backward passes reconstruct individual layer weights on the fly via ‘All-Gather’. Peak memory per GPU is:

$$M_{\text{ZeRO-3}} = \frac{2\Psi}{N_{dp}} + \frac{2\Psi}{N_{dp}} + \frac{12\Psi}{N_{dp}} \quad (63)$$

Additionally, **ZeRO-Offload** offloads partitioned states to host CPU memory, utilizing PCIe bandwidth to scale training to trillion-parameter models on limited GPU nodes.

Architecture Flow Diagram

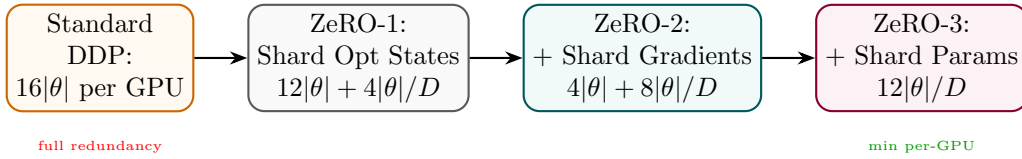


Figure 32: DeepSpeed ZeRO stages: each stage shards an additional component, progressively reducing per-GPU memory.

State Transition Table

Property	Standard DDP	ZeRO-3
Optimizer state memory	$8 \theta $ per device	$8 \theta /D$
Gradient memory	$4 \theta $ per device	$4 \theta /D$
Parameter memory	$4 \theta $ per device	$4 \theta /D$
Total per device	$16 \theta $	$16 \theta /D$
Max trainable size	GPU VRAM bound	$D \times$ larger

Table 32: Memory comparison between standard DDP and DeepSpeed ZeRO-3.